Using Weak Light Sources to Power Sensor Nodes for Sustainable IoT

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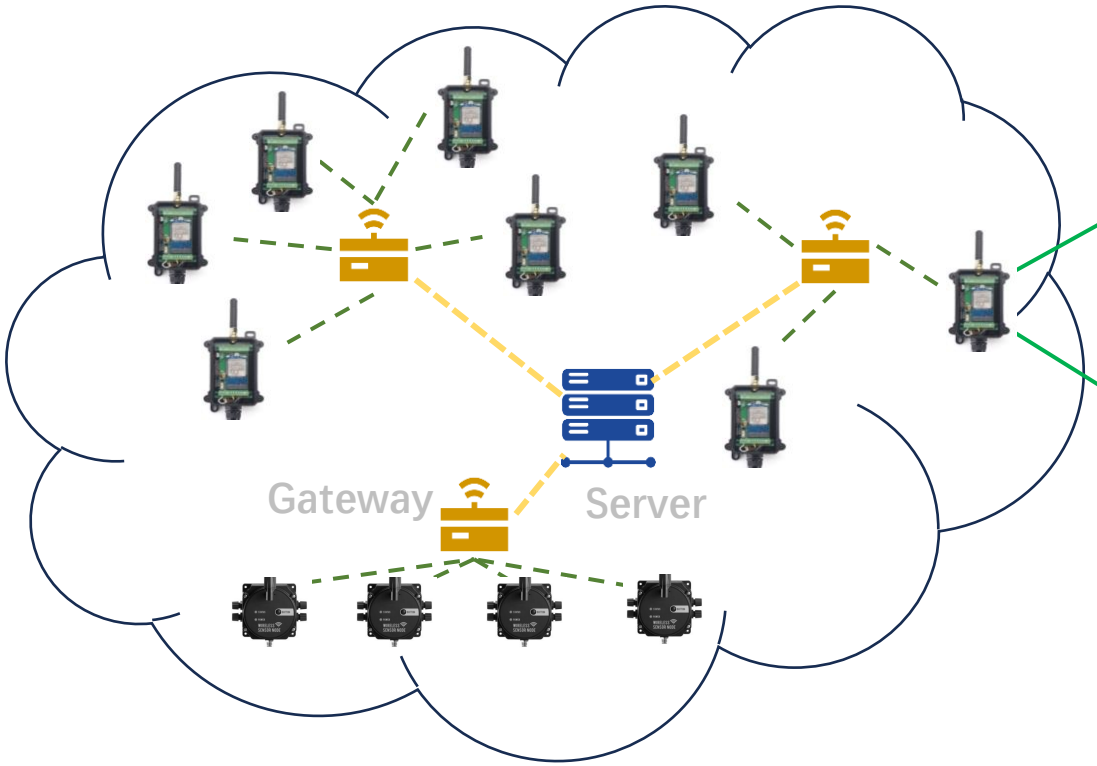
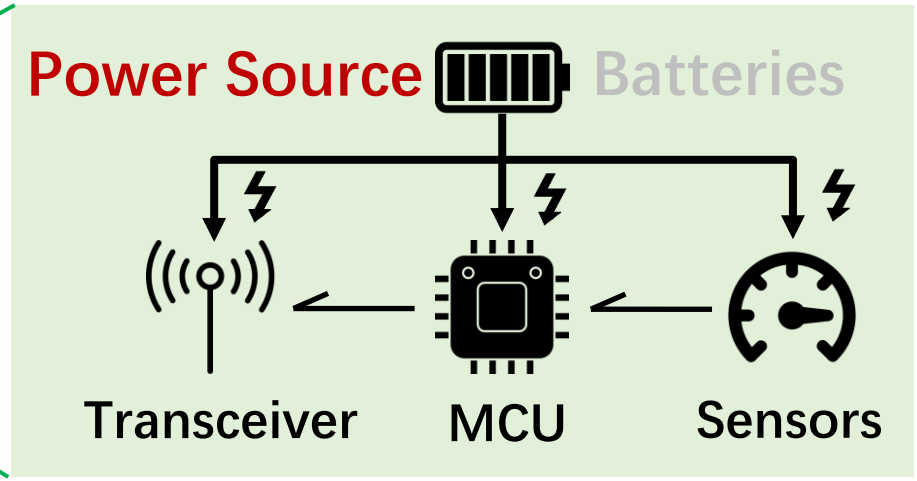
Background

◆Internet of Things (IoT) System

- Large number of IoT nodes (small sensing and communication devices)
- Critical pillar to build next-generation smart cities



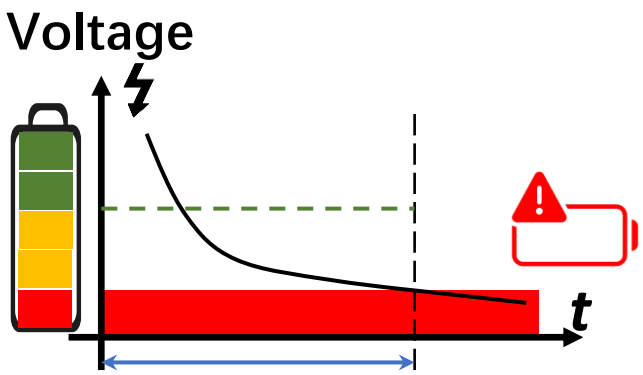
Key Components of IoT Node



Battery Power Limitation

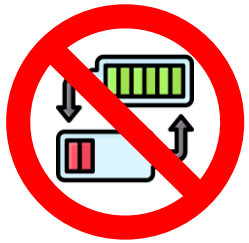
◆Existing Battery Power Sources Limit IoT's Wider Adoption

1) Short Working Lifespan



8~26 months only

2) Limited Deployment Flexibility

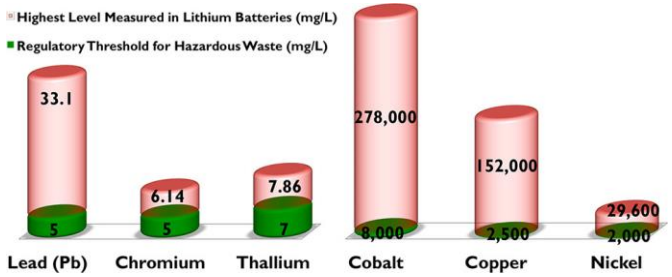


Difficult to replace batteries

3) Eco-Hostile

78 M/day

Used batteries



→ **Energy-Harvesting!**

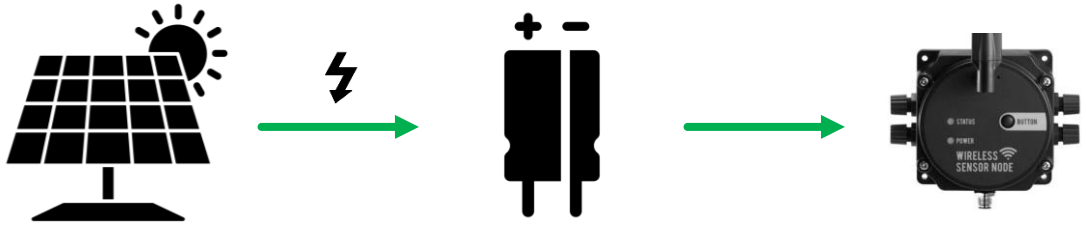
CORDIS, "Up to 78 million batteries will be discarded daily by 2025, researchers warn" Apr. 2024

D. Kang, et al, Potential Environmental and Human Health Impacts of Rechargeable Lithium Batteries in Electronic Waste. Environmental Science &Technology, 2013.

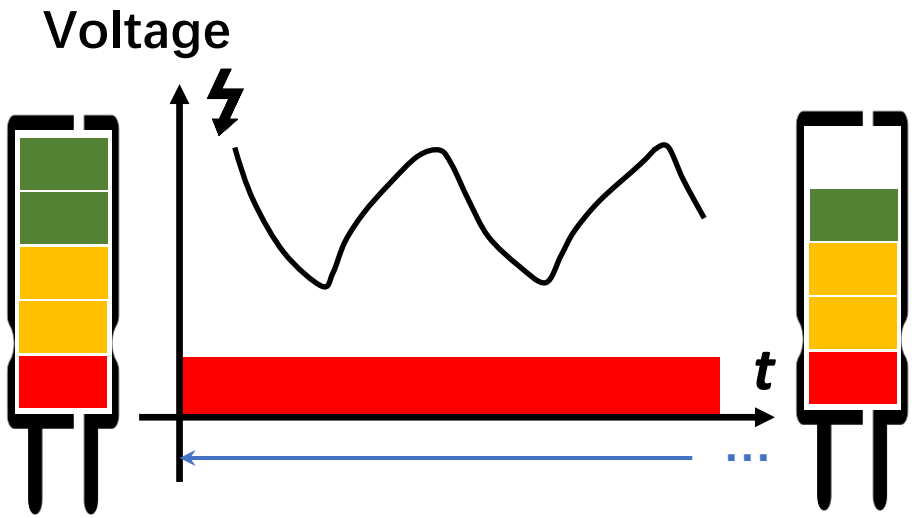
Why Energy-Harvesting?

◆Energy-Harvesting to Power IoT Nodes

Which energy source is most **practical and available**?



Power Source (e.g., light, thermal, etc.) → **Power Storage** (e.g., supercapacitor) → **Node Modules** (e.g., sensors, MCU, etc.)



Advantages:

1) **Self-powered & long lifetime**, e.g., 10~20 Years

2) **Flexible deployment**, e.g., no replacement for long

3) **Eco-friendly**, e.g., no harmful chemicals

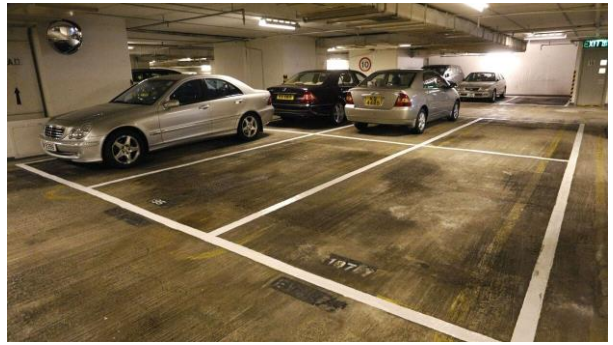
Weak Light Environment

Nodes can't get enough energy!

◆Light is Most Ubiquitous, but if Using It to Power IoT

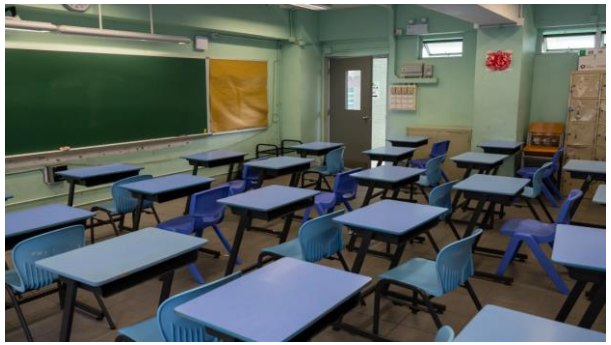
- Weak light sources: artificial light in buildings or blocked sunlight
- Nodes can only use weak light sources in many applications

1) Smart Home



2) Parking lot

3) Office



4) School

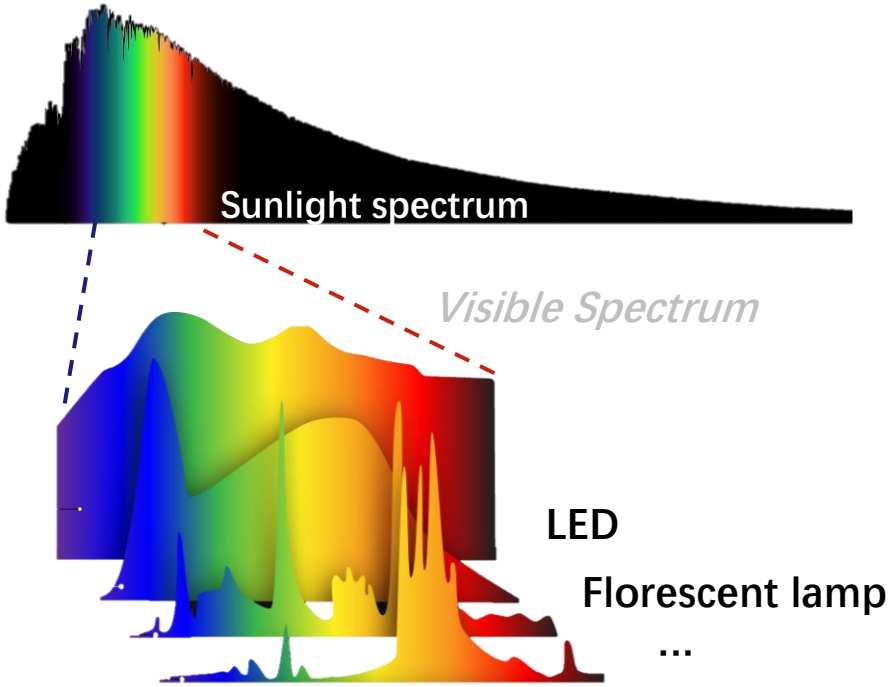
5) Shopping mall

6) Country park

Challenge

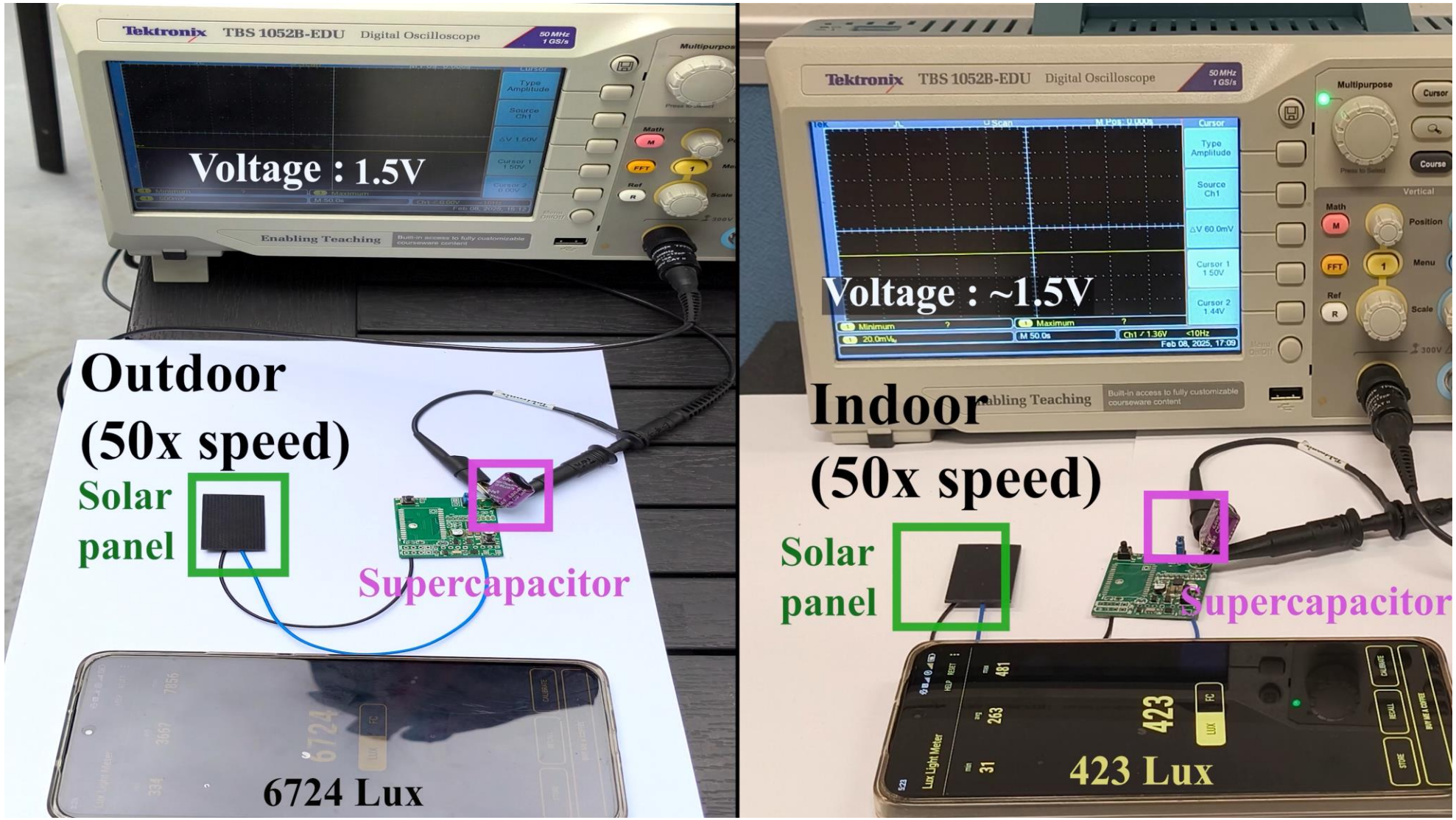
◆Why is Using Weak Light Sources Challenging?

1) Low energy harvesting efficiency



S. Peng, C.P. Low, Prediction free energy neutral power management for energy harvesting wireless sensor nodes,Ad Hoc Networks,2014

Weak Light Environment Demo



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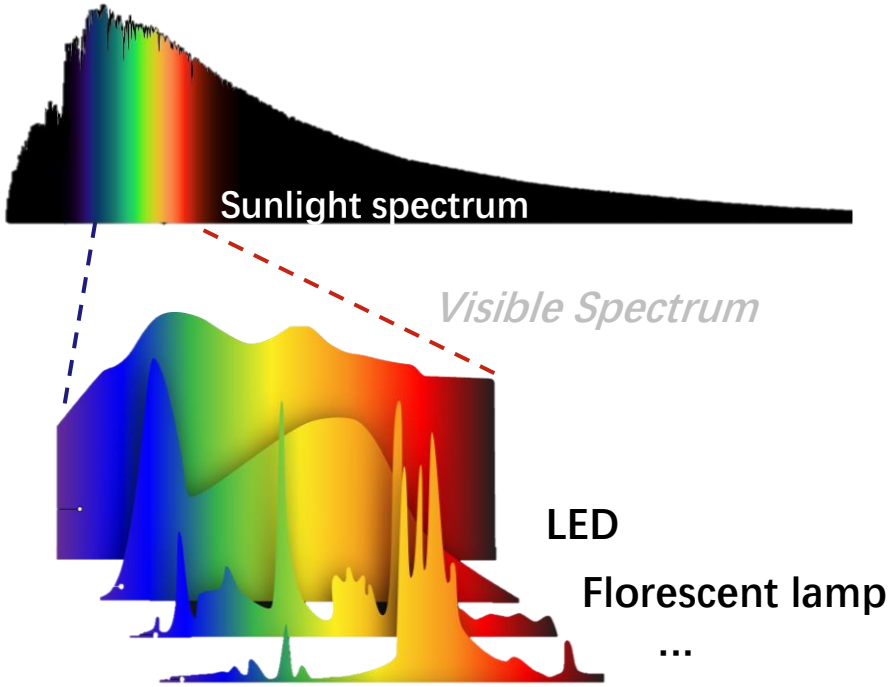
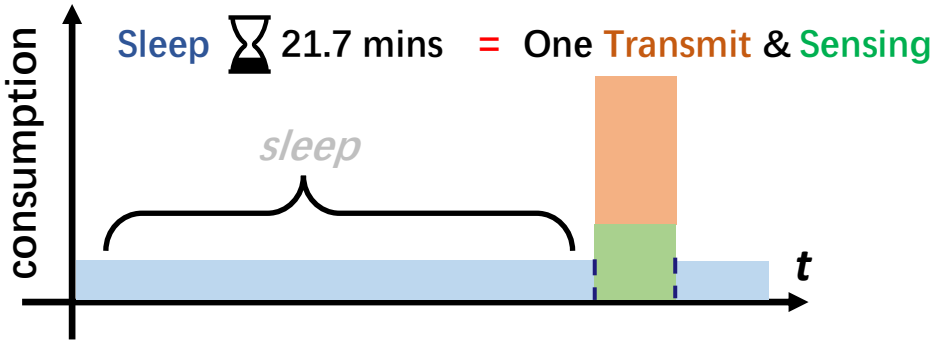
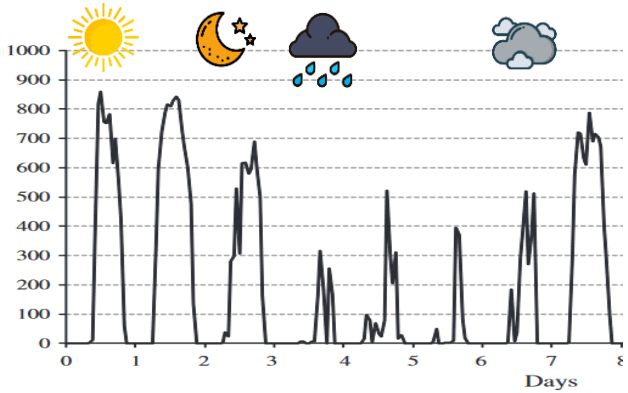


2) High power consumption in sleep state



3) Unstable energy income

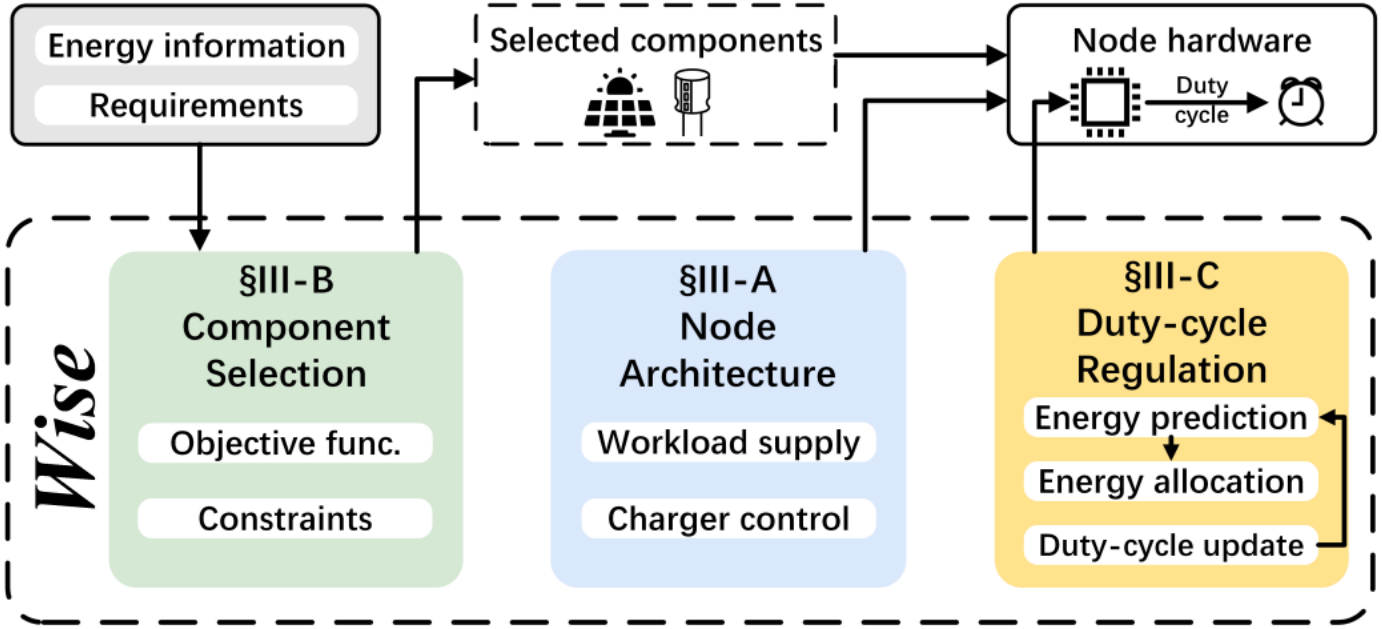


S. Peng, C.P. Low, Prediction free energy neutral power management for energy harvesting wireless sensor nodes,Ad Hoc Networks,2014

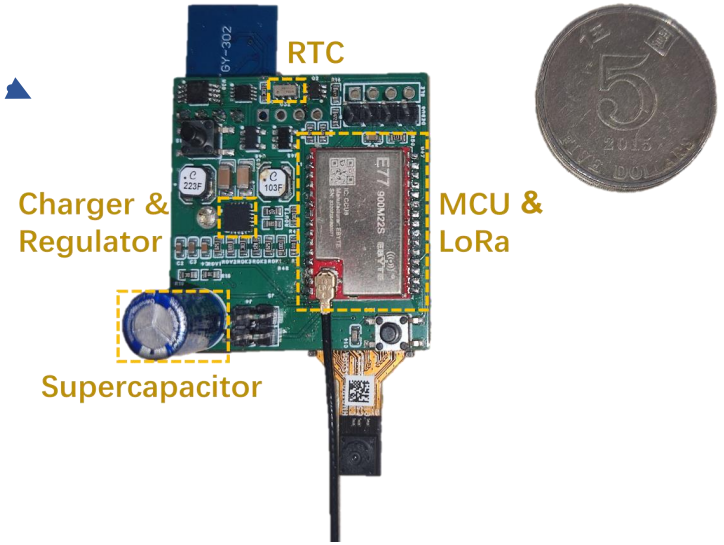
Overview

◆WISE: Use Weak Light Sources to Power IoT Node

- **Full-stack solution: encompasses both hardware and software designs**



Toolkit Hardware Driver

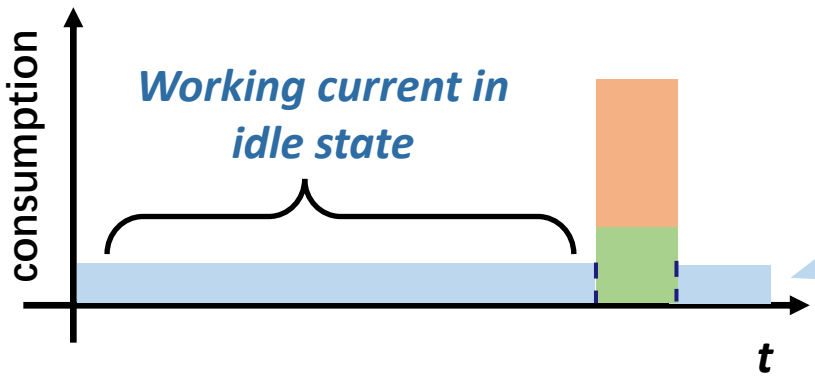


The prototype of our hardware

Observation

◆How High is the Current of IoT Nodes? And Why?

Quiescent current



Sleep leak **21.7** mins

= One Transmit & Sensing

Why High Quiescent Current?

- **MCU timer**
- ➔ Reused in sleep state
- ➔ For timekeeping and waking

Complex!

High power!

Vision

- **Simple function**
- ➔ Delegate to alternative low-power hardware.

MCU

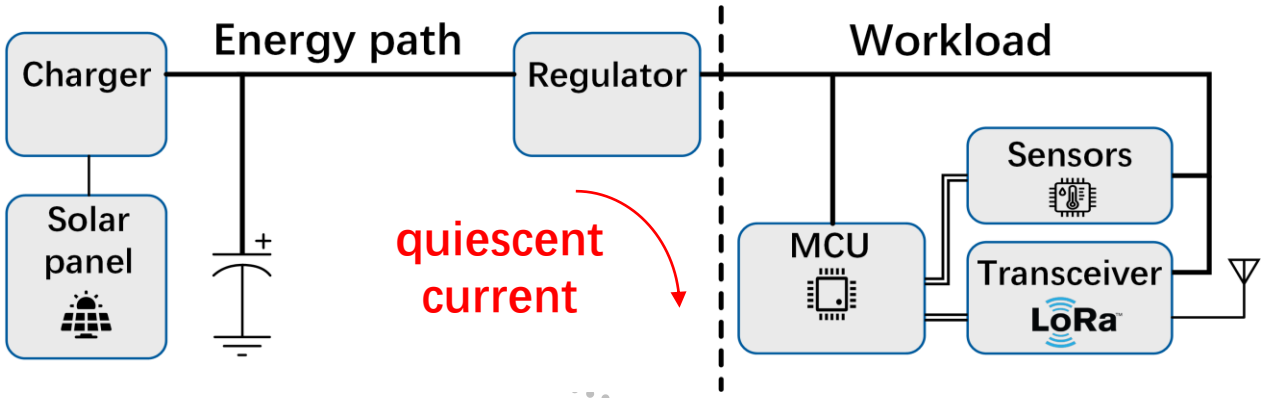
➔**Reduce *quiescent current!***

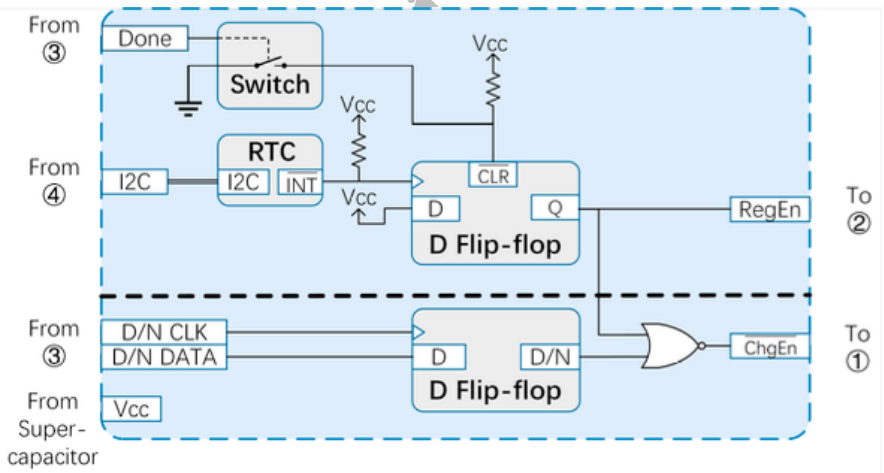
Node Architecture

◆Hardware Architecture

Key requirement: **ultra-low quiescent current** (i.e., current when node is idle)

Traditional Hardware Architecture





Key Design:

1: A dedicated timing **control part**

➔ **RTC**(Real-time clock)

➔ **ultra-low-power** timekeeping

2: Completely cut off **MCU** in sleep state

3: **Logic gates**

➔ shape the RTC signal

➔ Enable interaction with MCU and regulator

Achieve tens~hundreds times power reduction!

Duty-Cycle Regulation

◆Key Driver of the Novel Hardware

Dynamically adjust the working duty cycle according to the energy income

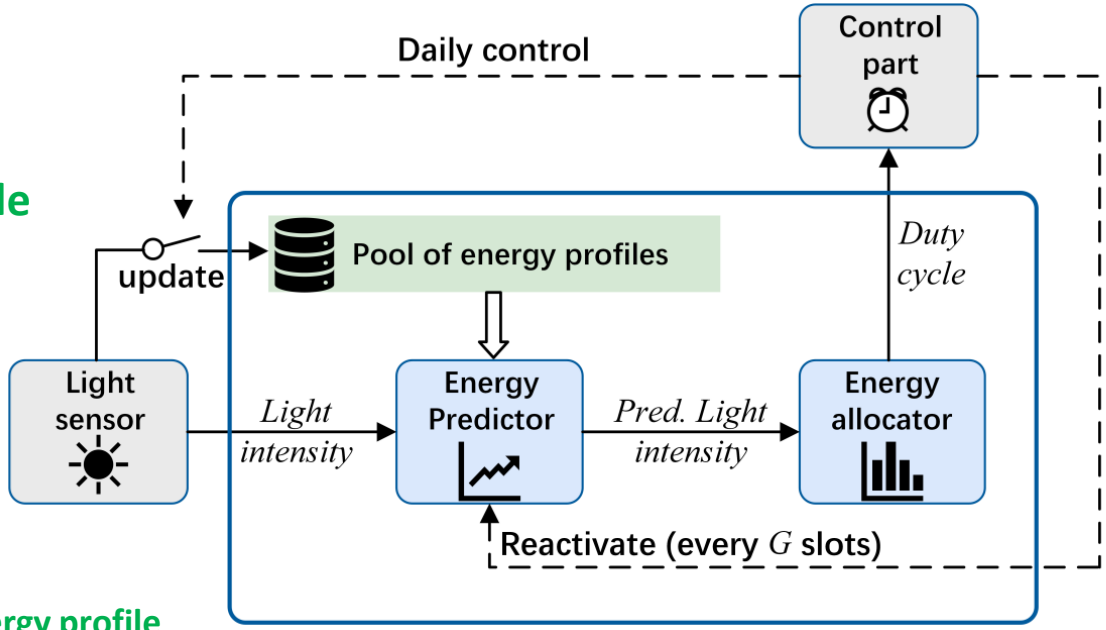
(sensing and communication frequency)

Energy predictor:

- Current energy income
- Historical energy profile

Weighted average

- Predicted energy income For rest of the day



Energy allocator:

- Current stored energy + Predicted energy sum

Energy budget

Update periodically

duty cycle

Current energy income historical energy profile

$$\hat{E}_{t+i} = \begin{cases} \gamma_i \times E_t + (1-\gamma_i) \times E^f_{t+i}, & \text{if } i \le G, \\ E^f_{t+i} & \text{if } i > G. \end{cases}$$

$$E_{total} = E_t^{sc} + \sum_{j=1}^{N-t} \hat{E}_{t+j}$$

Current stored energy predicted energy sum

Component Selection

◆ Component Selection Toolkit

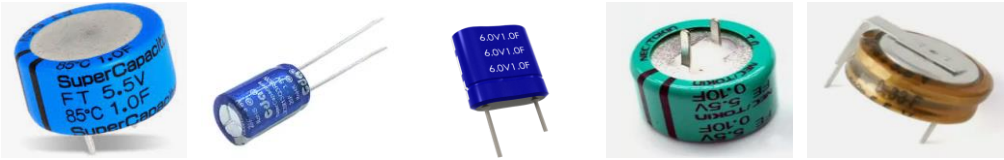
- For each component, there are many products on the market
- Good products are usually expensive -> trade-off of performance & cost
- Toolkit suggests the optimal component combination based on budget

Solar panel

E_{in}

Income energy

Super-capacitor



E_{sc}

Stored energy

$\sum_i c_{p_i}$: Costs of different components

cost

performance

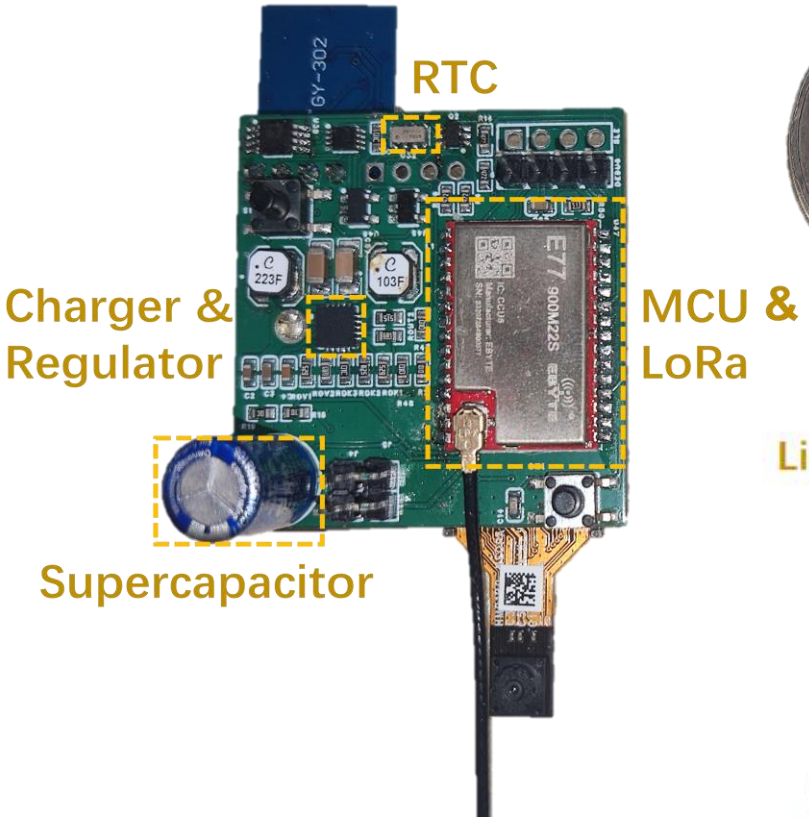
$$\arg\max_{\mathrm{p}}\left(\frac{w}{\sum_i c_{p_i}} + (1-w)\times r_{\mathrm{p}}\right),$$

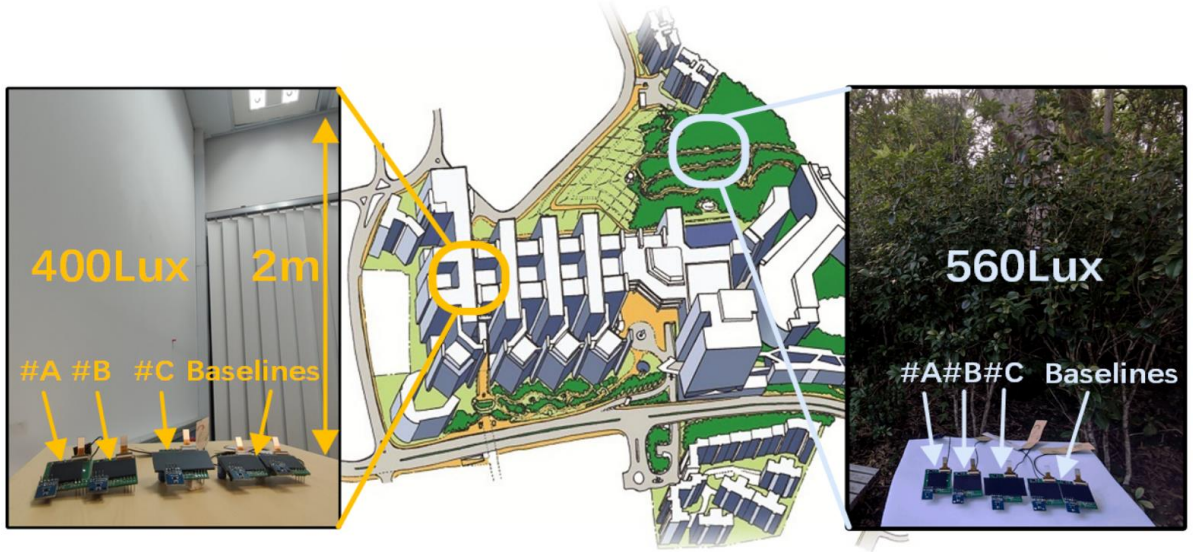
$$s.t. \quad r_{\mathrm{p}} > 1, \left(r_{\mathrm{p}} := \frac{E_{in_{\mathrm{p}}}}{E_{con_{\mathrm{p}}}}\right)$$ *Energy income*

$$\sum_i c_{p_i} < b,$$ *Cost budget*

$$E_{sc} > E_{in}$$ *Energy storage*

Implementation





DEFAULT LORA PARAMETERS USED IN THE EVALUATION.

SF	BW	CR	f_c	P_{Tx}	Preamble	Data length
8	125 KHz	4/5	915.125 MHz	10 dBm	20 Symbols	25 Bytes

Charger & Regulator	BQ25570
RTC	RV3028-C7
MCU & LoRa	STM32WLE5CCU6/ E77-900M22S
Camera	HM01B0
Light sensor	BH1750

Evaluation

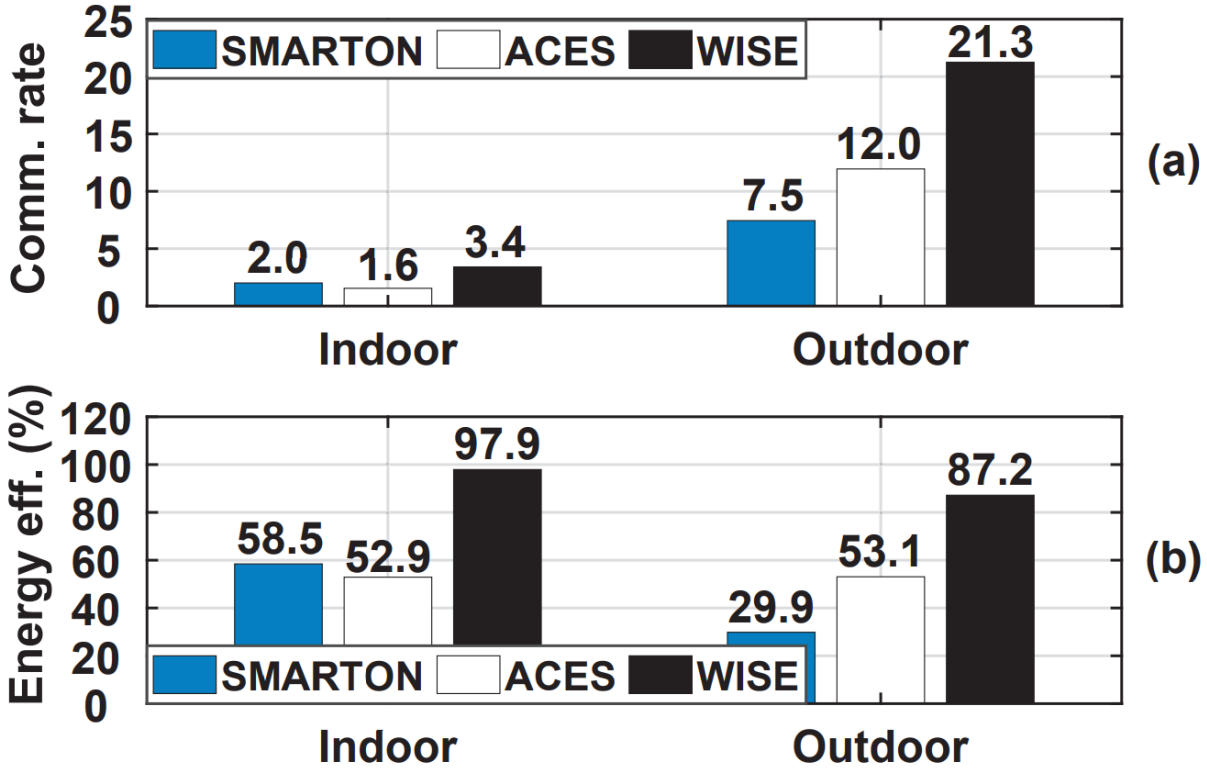
Indoor scenarios:

Comm. rate:

69.7% -- 120.0%

Energy eff.:

67.5% -- 85.1%



Outdoor scenarios:

Comm. rate:

77.8% -- 185.2%

Energy eff.:

64.3% -- 192.0%

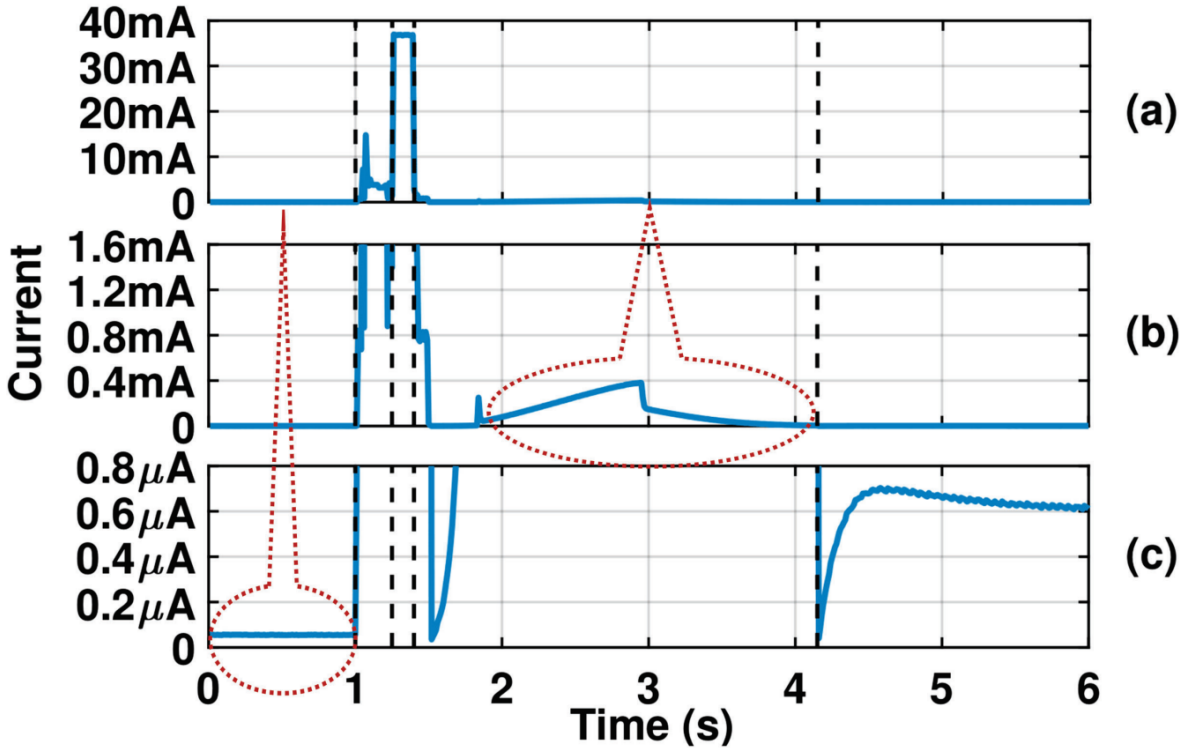
Overall performance

F. Fraternali, et al. Aces: Automatic configuration of energy harvesting sensors with reinforcement learning. ACM Transactions on Sensor Networks, 2020.

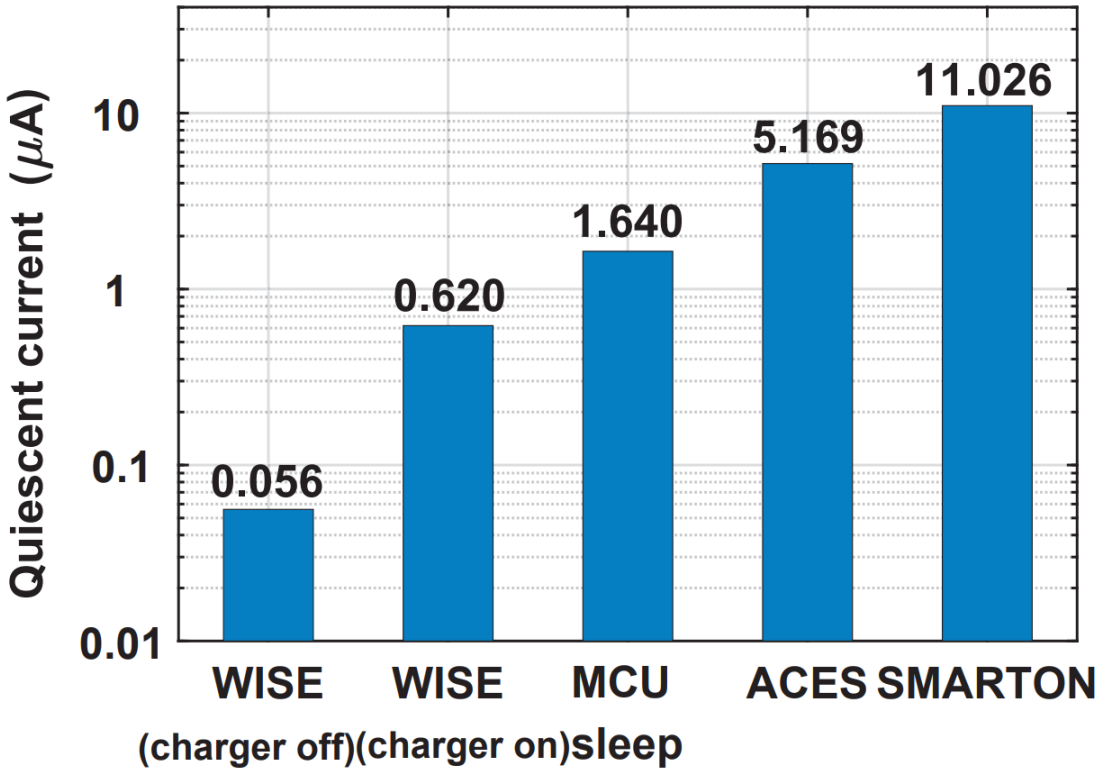
Y. Luo and S. Nirjon. Smarton: Just-in-time active event detection on energy harvesting systems. In Proc. of IEEE DCOSS, 2021.

Evaluation

Quiescent current reduction: ***88.0–99.5%***

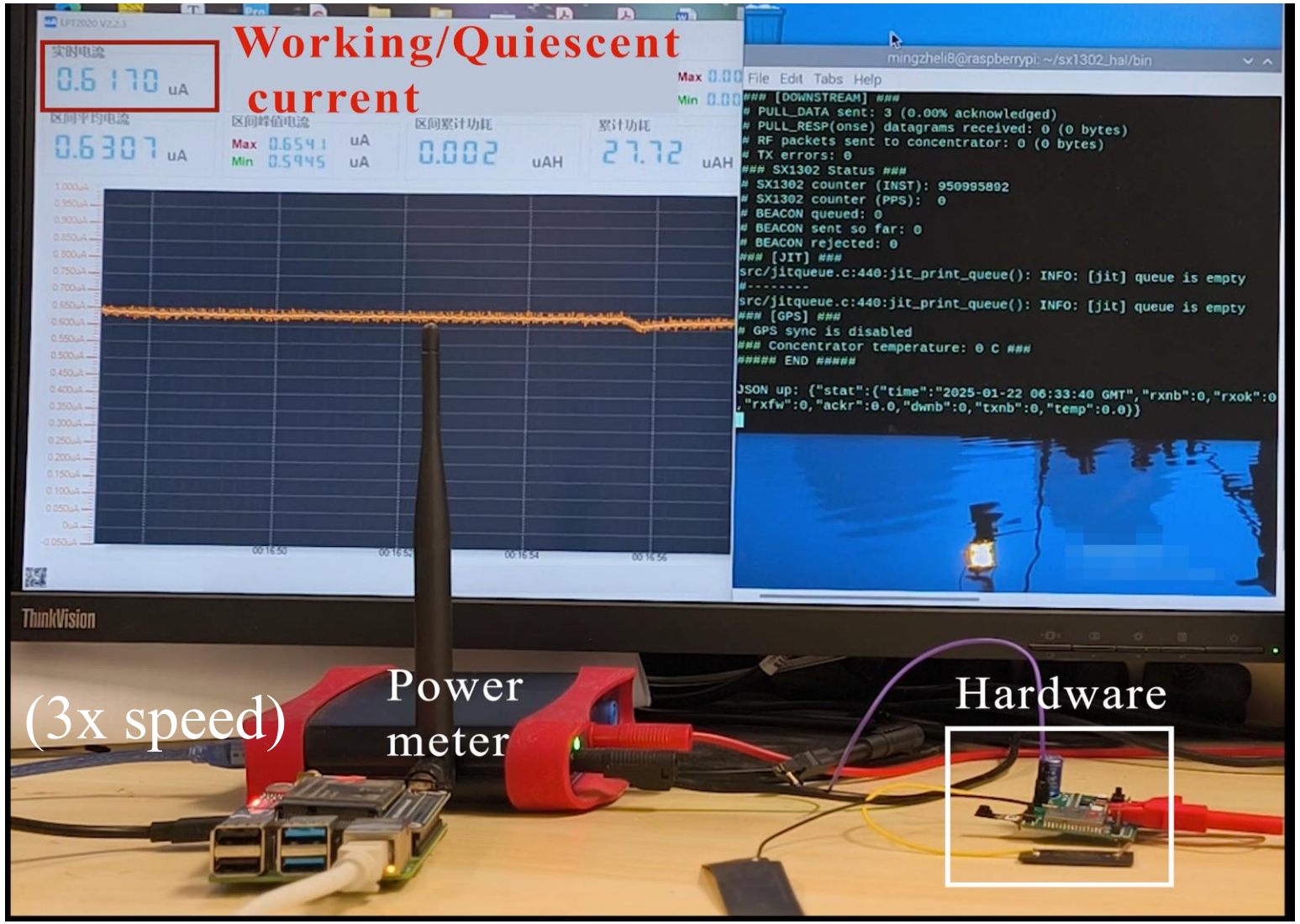


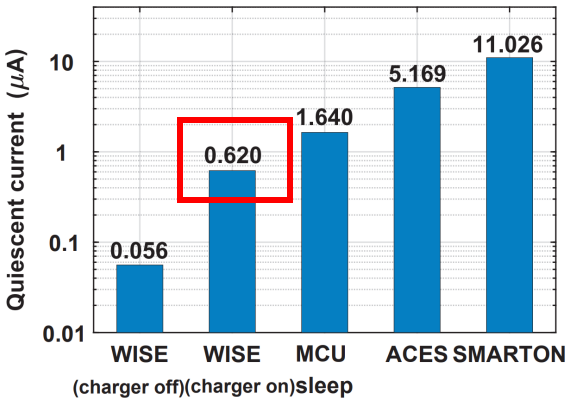
Runtime current



Quiescent current comparison

Current Demo





Microbenchmark

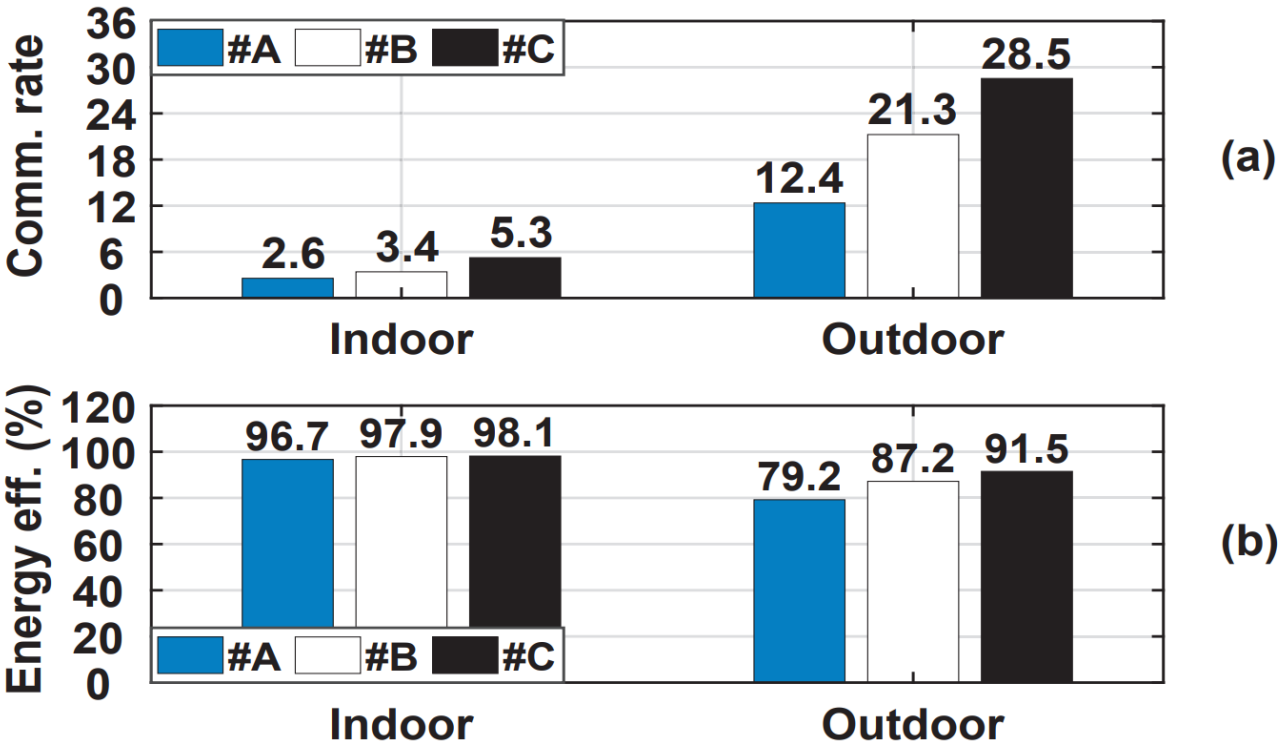
SOLAR PANELS AND SUPERCAPACITORS SELECTED AT EACH NODE.

#	Solar panel	Supercapacitor	Cost (USD)	r_p
A	KXOB081K06TF (23mm×25mm)	LIC0813Q3R8106 (10 F)	8.89	1.3674
B (Default)	SM141K05TF (22mm×35mm)	LIC0813Q3R8206 (20 F)	9.47	1.8315
C	SM141K07TF (23mm×50mm)	LIC0820Q3R8406 (40 F)	11.84	2.7348

#A: Prefer low cost

#B: Balanced

#C: Prefer performance



Overall performance of different implement set

Component Selection Toolkit Balanced node cost and performance

Take Away

- We introduce WISE, a sustainable IoT platform can be **powered by weak light** environments
- We propose a new IoT node design with an **ultra-low power architecture** that **reduces the quiescent current** by ***88.0–99.5%***
- We also design an optimization scheme to **select energy-related components** and an **energy-aware duty cycle driver** to operate WISE platform

Thank you!
Q&A

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